Aim:- Study of various Network comm
Linux and Windows.

BASIC Networking Commands :-

1) arp -a :- ARP is short form of address resolution Protocol, it will show the IP address of your computer. along with IP address and MAC address of your router

- gateway (10.0.2.2) at 52:55:0a:00:02:02 [ether] on enp0s3.

2) hostname - this is the simplest of all TCP/IP Commands. It simply displays the name of your computer.

- fedora

3) ipconfig /all :- this command displays detailed Configuration information about your TCP/IP connection including Router, Gateway, DNS, DHCP, and type of Ethernet adapter in your system.

4) nbstat -a :- this command helps solve problems With NetBIOS name resolution

5) Physical address - F8-ED-FC-5C-0A-00.

6) IPV4 Address - 172.16.10.150

7) Subnetmask - 255.255.25.0

8) DHCP Enabled - NO

9) netstat :- displays varity of statistics about a computers active TCP/IP connections It is a command line tool for monitoring network connections both incomming and outgoing.

- Active Internet connections (w/o servers)

Prto	Recv-Q	Send-Q	Local Address	Foregin Address
Udp	0	0	fedora:bootpc	-gateway

↳ Fedora

9) nslookup :- A Tool used to perform DNS Lookups in Linux. Used to display DNS details like IP address of a specific computer, MX records for a domin (or) NS servers of a domain. nslookup can operate in 2 mode interactive and non-interactive.

Eg:- nslookup www.google.com.

O/P:-

~~Default~~ Server :- ~~Unknown~~ 127.0.0.53

Address : fe80::b0be:89ff:fe32:8c64

↳ CP

10) Pathping - Pathing is unique to window's and is basically a combination of Ping and Tracert Commands. Pathping traces the route to destination address then launches a 2.5 second test of each router along the way, gathering statistics on the rate of data loss along each hop.

O/P:-

Usage : pathping ~~[-n DOMAIN]~~ [-g host-list] [-h maximum_hops] [-i address] [-n] [-p period] [-q num_queries] [-w timeout] [-4] [-6] target_name.

11) Ping - Packet INternet Groper command it tests Connectivity btw two nodes. Ping use ICMP to communicate to other devices.

Usage: ping [-t] [-a] [-n count] [-l size]
[-i TTL] [-v TOS] [-r count] [-s count]
-J host-list] | [-k host-list]] [-w timeout]

Route- It shows / manipulate IP routing table.
It is primarily used to setup static routes
specific host/ networks using an interface.

O/P:-
Manipulates network routing tables.
ROUTE [-f] [-p] [-4|-6] command [destination]
[MASK netmask] [gateway] [METRIC metric]
[IF interface]-

me Important Linux networking commands:-
1) ip - it set's up new saystem
nd assigining IP's to trouble shooting existing
system. ip command can show address
information, manipulate routing, plus display
network various devices, interfaces and
tunnel.

1) #Ping www.google.com

O/P:- Pinging www.google.com [142.251.151.119]
With 32 bytes of data:
Reply from 142.251.151.119: bytes=32
time=5ms TTL=119.

2) # ping ip address (ping 4.2.2.2)
O/P:- Ping 172.16.8.1
Pinging 172.16.8.1 with 32 bytes of data:
Reply from 172.16.8.1: bytes=32 times<1ms TTL

3) #ping fully qualified

o/p:-

www.facebook.com.

Pinging star-mini.clor.facebook.com
[57.144.56.1] with 32 bytes of data

Reply from 57.144.56.1: bytes=
time=4ms TTL=55

a. To show the IP address assigned to an interface on your server:-

a. [root@server ~]# ip address show

1: lo: <LOOPBACK, UP, LOWER_UP> mtu 65536
qdisc noqueue state unknown group default
qlen 1000 link/loopback 00:00:00:00:00:00
brd 00:00:00:00:00:00.
inet 127.0.0.1/8 scope host lo.

b. To assign an IP to an interface, for example, enps03:

a. [root@Server ~]# ip address add
192.168.1.254/24 dev enps03

1: lo: <LOOPBACK, UP, LOWER_UP> mtu 65536 qdisc
noqueue state unknown group default qlen 1000
inet 127.0.0.1/8 scope host lo.

c. To delete an IP on an interface

a. [root@Server ~]# ip address del
192.168.1.254/24 dev enps03.

ter the status of the interface by bringing
e interface eth0

Online: [root@server ~]# ip link set eth0 down

npOS3: <BRODCAST, MULTICAST> mtu 1500 qdisc
l state DOWN group default qlen 1000 link/ether
27:fc:f3:9b brd ff:ff:ff:ff
me enx080027fcf39b.

ter the status of the interface by bringing
the interface eth0.

offline: [root@server ~]# ip link set enp0s3
Promisc on

npOS3: <BRODCAST, MULTICAST, PROMISC> mtu 1500
fq_codel state DOWN group default qlen 1000
ther 08:00:27:fc:f3:9b.

dd a default route (for all addresses) via
e local gateway 192.168.1.254 that can
reached on device eth0:

#ip route add default via 192.168.1.254
dev ~~eth0~~ enp0s3

pOS3:- <BRODCAST, MULTICAST, PROMISC, UP, LOWER_UP>
1500 qdisc fq_codel state up group default qlen
altname enx080027fcf39b.

d a route to 192.168.1.0/24 via the
ateway at 192.168.1.254:

oot@server] #ip route add 192.168.1.0/24
via 192.168.1.254

:-
RTNETLINK answers: File exists

dd a route to 192.168.1.0/24 that can be
ched on device.

eth0: #ip route add 192.168.1.0/24 dev
eth0.

j. Delete the route for 192.168.1.0/24 via the gateway at 192.168.1.254:

```
#ip route delete 192.168.1.0/24 via
192.168.1.254.
```

O/P:-

RTNELINK answers: No such Process,

K. Display the route taken for IP 10.10.1.4:

```
#ip route get
10.10.1.4
```

O/P:-

10.10.1.4 dev enp0s3 src 192.168.1.254
Vid 0 cache.

2) ifconfig- Command was/is a staple in many sysadmin's tool belt for configuring and troubleshooting networks. It has since been replaced by ip command.

O/P: -

enp0s3 : flag = 4419 <UP, BROADCAST, RUNNI
PROMISC, MULTICAST> Mtu 1500
inet 192.168.1.254 netmask 255.255.255.
broadcast 0.0.0.0
ether 08:00:27:fc:f3:9b txqueuel
1000 (Ethernet)

r - Matt's traceroute is a program
a command-line interface that serves
network diagnostic and troubleshooting
ol. this command combines the funcitonality
of the ping and traceroute commands.
like a traceroute, mtr command will
w the route from a computer to a
d host. mtr provides a lot of stastics
ut each hop, such as response time and
cage. get more into about the route and
able to see problamatic device along the way.
see a sudden increase in response time
packet loss, then obi, there is a bad link
mewhere.

ax:-

mtr <options> hostname/IP

asic mtr command shows you the statistics
ding each hop (hostnames) with time and loss %:

mtr google.com

	Loss%	Snt	last	Avg	Best
st	0.0%	10	0.6	0.7	0.5
teway.local	0.0%	10	2.1	2.3	2.0
0.10.0.1					
sp-router.example.net	0.0%	10	8.5	8.9	8.2
orel.isp.net	0.0%	10	15.4	15.8	15.0

how numeric IP address (i) you use -g, you
ill get IP address (numbers) instead of
ostnames):-

t # mtr -g google.com

	Loss%	Snt	Last	Best	Wrst
192.168.1.1)	0.0%	10	0.6	0.5	1.2
10.10.0.1)	0.0%	10	2.1	2.0	3.1
203.0.113.1)	0.0%	10	8.5	8.2	10.1
198.51.100.10	0.0%	10	15.4	15.0	17.2

c. Show the numeric IP address and hostnames, too:

#mtr -b google.com

O/P:-

Host	Loss%	Snt	Last
1) gateway.local	0.0%	10	0.05
2) 10.10.0.1	0.0%.	10	0.05
3) isp-router.example.cnet	0%.0%.	10	0.05

d. Set the number of pings that you want to send:-

mtr -c 10 google.com.

Start:- 2020-07-28T12:00:00+0000
Host:- Server.Example.com

	Loss%	Snt	Last
1. gateway.local			
2. 10.0.0.1			
3) isp-router.example.net			
4) corel isp.net			

4. tcpdump - is designed for capturing and displaying packets.

dnf install -y tcpdump.

Before, starting any capture you need to know which interfaces tcpdump can use. You will need to use sudo or have root access in this case.

tcpdump -D

If you want to capture traffic on eth0, you can initate that with tcpdump eth0 sample

O/P:-

#tcpdump -i eth0.

#tcpdump -i etho -c 10

Capture traffic to and from one host.
can filter out traffic coming from a
special host. For eg:-, to find coming
and going to 8.8.8.8, use the command:

#tcpdump -i etho -c 10 host 8.8.8.8

raffic comming from 8.8.8.8 use:-

#tcpdump -i etho src host 8.8.8.8

O/P:-

12:55:01.123456 IP 192.168.1.10.5300
> 8.8.8.8.53: DNS query

HTTPS Traffic with host 8.8.8.8

/P:-

12.55.01.123456 IP 192.168.1.10.53000
> 8.8.8.8.53 DNS query

~~12:55~~

HTTPS traffic to google

O/P:- 13:00:01.123456 IP

Ping:-

Ping is a tool that verifies IP-level
nnectivity to another TCP/IP computer by
ending Internet Control Message Protocol
ICMP) Echo request messages. the recipt
of corresponding Echo Reply messages
s displayed along with round-trip times.
ing is the primary TCP/IP command
used to troubleshoot aconnectivity,
reachibility, and name resolution.

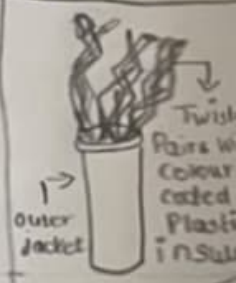
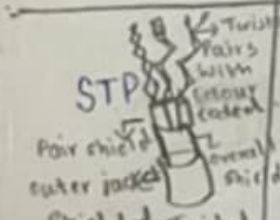
Practical-2

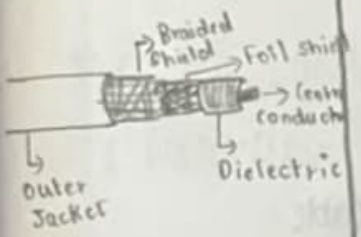
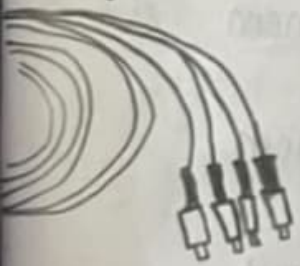
Aim:- Study of different types of Network Cables.

a) Understand different types of network cable.

Different type of cables used in networking are:-

1) Unshielded Twisted Pair (UTP) Cable.
2) Shielded Twisted Pair Cable (STP)
3) Coaxial Cable
4) Fibre Optic Cable.

Cable Type.	Category	Maximum Data Transmission	Adv / DisAdv	Application/ Use
UTP	Category 3	10 bps	Adv:- • Cheaper in cost • Easy to install as they have a smaller overall Diameter. DisAdv:- • More Prone to (EMI) Electromagnetic interference and noise	10 Base-T Ethernet
	Category 5	up to 100 Mbps		Fast Ethernet, Gigabit Ethernet
	Category 5e	1 Gbps		
STP  Shielded Twisted Pair SSTP	Category 6, 6a	10 Gbps	Adv:- • Sheilded • Faster than UTP • Less Susceptible to noise and interference Dis:- • Expansive • Greater installation	Gigabit Ethernet 10G Ethernet (55 m) Widely used in data Centres Gigabit Ethe 10G Ele me (100 m)
	Category 7.	10 Gbps		
				

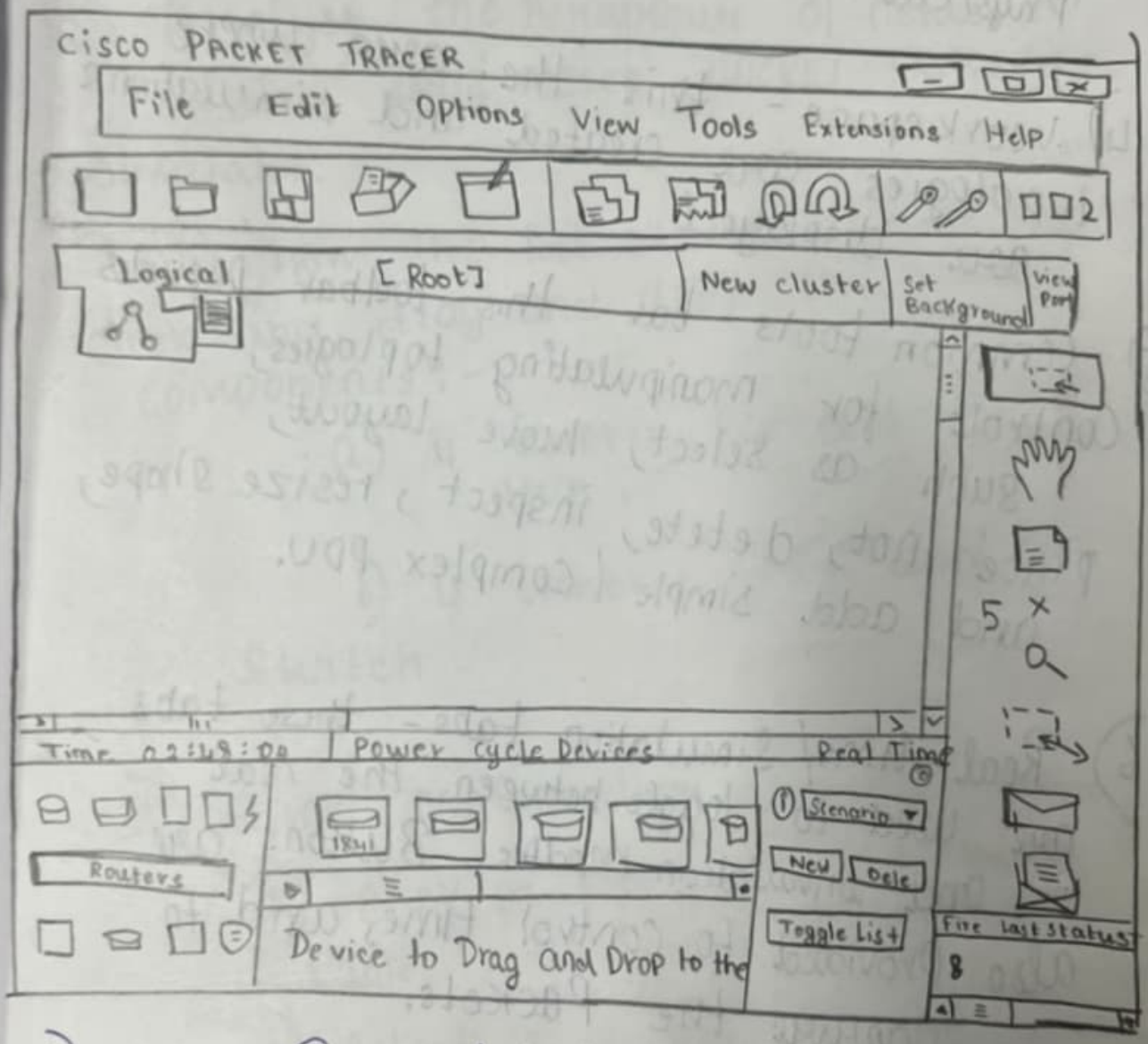
Coaxial Cable image :-  	RG-6 RG-59 RG-11	10-100 Mbps	• High Bandwidth • Immune to interference • Low loss bandwidth • Versatile • DisAdv:- • Limited Distance • Cost • Size is bulky	Speed of Signal is 500m Television network High speed internet Connection
Fibre Optics Cable image :- 	Single Mode Multi Mode	100Gbps	Adv:- • High speed • High Bandwidth • High Security • Long distance DisAdv:- • Expensive • Requires skilled installers	• Maximum distance of Fibre optics cable is around 100 metere.

b) Make your own Ethernet Cross-Over Cable / Stright Cable :-

Tools and parts needed :-

- Ethernet Cabling CAT5e is Certified for gigabit support, but CAT5 Cabling works as well, just over short distances.
- A Crimping tool. this is an all-in-one networking tool shaped to push down the pins in the Plug and strip and cut the shielding OH the Cables.

the Layout of Packet Tracer is divided into several components. the components of the Packet Tracer Interfaces are as follows: Math the numbering with explanations.



1) Menu Bar - this is a common menu found in all software applications. it is used to open, save, Print, change Preferences and so on.

2) Main toolbar - this bar provides shortcut icons to menu options that are commonly accessed, such as open, save, zoom, undo, redo.

3) Logical / Physical Workspace tabs
— these tabs allow you to toggle between the Logical and Physical work areas.

4) Workspace - this the area where topologies are created and simulations are displayed.

5) Common tools bar - this toolbar provides Controls for manipulating toplogies, such as Select, Move layout, place not, delete, inspect, resize shape, and add simple / complex PDU.

6) Real-time/ Simulation tabs - these tabs are used to toggle between the real and simulation modes. Buttons are also provided to control time, and to capture the Packets.

7) Network component box - this Component contains all of the network and end devices available with Packet Tracer, and is further divided into two areas : Area 7a : Device-type ction box -

ng google.com
, google.com (216.58.206.174) 56(84) bytes of
a.
bytes from sof02s227-in-f14.

guring an Ethernet connection by using nmeli
if you connect a host to the network
ver Ethernet, you can manage the connection's
gs on the command line by using the nmeli
tility.

edure:-
List the Network-Manager connection
Profiles:-
nmcli connection show
VAME UUID TYPE DEVICE
ed connection 1 a5eb6490-cc20-3668-8148-03
ethernet enp1s0

nmcli, Display the current settings
the connection.
Profile: # nmeli connection show
connection.interface-name:
enp1s0
connection.autoconnect yes
ipv4.method: auto
ipv6.method: auto

5. Configure the IPV4 settings:

- To Use DHCP, enter:

nmeli connection modify "Wired connection 1" ipv4.method auto.

- To set a static IPV4 address, network mask, default gateway, DNS servers, and search domain, enter:

nmelic connection modify "Wired connection 1" IPV4.method manual IPV4.addresses.

6. Configure the IPV6 Settings:-

- To use stateless autoconfiguration enter:

nmeli connection modify "Wired Connection 1" IPV6.method auto

7. Activate the profile:-

nmeli connection up Internal-LAN

Verification

1. Display the ip settings of NIC:

ip address show enp1s0

enp1s0: <BRODCAST, MULTICAST, UP, LOWER_UP>
mtu 1500 qdisc.
fq-codel state UP group default
qlen 1000.

2) Display the IPV4 default gateway
ip route show default
default via 192.0.2.254 dev enp1s0 Proto static metric 102

3) Display the IPV6 default gateway
ip-6 route show default
default via 2001:db8:1::ttee dev enp1s0 Proto static metric 102 pref medium

4) Display the DNS settings:
cat /etc/resolv-conf
search example.com
nameserver 192.0.2.200
nameserver 2001:db8:1::tbb
if multiple connections profiles are Active at the same time, the order of nameserver entries depnd on the DNS priority values in these profile and the connection types.

5) Use the ping utility to verify that this host can send packets to other hosts:
ping <host-name-or-IP-address>

Student Observation :-

1) Which command is used to find the reachibility of a host machine from your device?

Command :- ping

o/p :-

Ping google.com

2) Which command gives the details of hops taken by a packet to reach its destination?

Command :- traceroute (Linux) or traceroute (Windows).

Eg :-

traeoute google.com

3) Which command displays the IP configuration of your machine?

Linux :-

ip addr show

(or)

ifconfig.

4) Which command displays the TCP port status in your machine?

Command :-

netstat -tuln

or

ss -tuln.

5) Write the command to modify the IP configuration in a Linux machine.

→ To assign a new IP address: -

sudo ip addr add 192.168.1.100/24 dev eth0

→ To Replace/change existing IP: -

sudo ip addr flus dev eth0

sudo ip addr add 192.168.1.100/24 dev eth0.

→ To Bring the interface up: -

sudo ip Link set eth0 up

- Two RJ45 plugs
- Optional two plug shields.

Step 1:- To start construction of the device, begin by threading shields onto the cable.

Step 2:- Next, strip approximately 1.50m of cable shielding from both ends. the crimping tool has a round area to complete this task.

Step 3:- After, you will need to untangle the wires there should be four "twisted pairs". Refreencing back to the sheet, arrange them from top to bottom. One end should be in arrangment A and the other in B.

Step 4:- Once the order is correct, bunch them together in a line, and if there are any that stick out farther than others, snip them back to create an even level. the difficult aspect is placing these into RJ45 Plug without messing up the order. To do so, hold. plug with the clip side facing away from you and have the gold pins facing towards you.

Step 5:- Next, push the cable right in. the notch at the end of the plug needs to be just over the cable shielding, and if it isn't that means you stripped off too much shielding. Simple snip the cables back a little more.

Step 6:- After the wires are securly sitting inside the plug, insert it into the crimping tool and push down.

Step 7:- Lastly, repeat for the other end using diagram B.

Student Observation:-

1) What is the difference between a Cross cable and a stright Cable?

→ Stright Cable - Both end uses the samre wiring standard (T568A - T568A or T568B - T568B) It is used to connect different types of devices.

→ Cross Cable - One end uses T568A and other uses T568B. it is used to connect Similar types of device.

2) Which Type of cable is used to Connect two PCs?

→ Cross Cable.

3) Which type of Cable is used to comect a router/switch to your PC?

→ Stright Cable

4) Find out the category of twisted pair Cable used in your lab to connect the PC to the network socket.

Ans → Category 6 (Cat6).

5) Write down your understanding, Challanges faced, and output received while making a twisted pair cross/ straight cable.

→ Understanding :-

→ i learned the difference btw Stright and cross cables.

→ i understand the T568A and Wiring standards.

→ i learned how to arrange wires correctly and crimp an RJ45 Connector.

Challanges faced :-

→ Arranging the wire colors in crt order.

→ Keeping wires stright before inserting them into RJ45 connector.

→ Ensuring all wires were fully inserted before crimping.

→ Verifying the cable with Cable tester.

Output recived :-

→ Succeffully prepared a stright/ Cross twisted pair cable.

→ the cable passed the cable tester and established a successful network connection btw the devi...

Date:- 3/08/2026 Practicle - 3.

Aim:- To study the Packet tracer tool Instalation and user Interface overview.

c) To understand environment of CISCO PACKET TRACKER to design simple network.

INTRODUCTION:-

A Simulator as the name suggests, simulates network devices and its environment. Packet Tracer is an exciting network design, simulation and modeling tool.

1) It allows you to model complex systems without the need for dedicated equipment.

2) It helps you to practice your network configuration and troubleshooting skills via computer (or) an Andorid or ios based mobile device.

3) It is avialable for both Linux and windows desktop environment.

4) Protocols in Packet Tracer are coded to work and behave in the same way as they would on real hardware.

8) User-Created Packet box - Users can create highly-customized packets to test their topology from this area, and the results are displayed as a list.

d) Analyse the behaviour of network device using CISCO PACKET TRACER Simulator.

1) From the network component box, click and drag-and-drop the below components:-

a) 4 Generic PCs and one HUB.

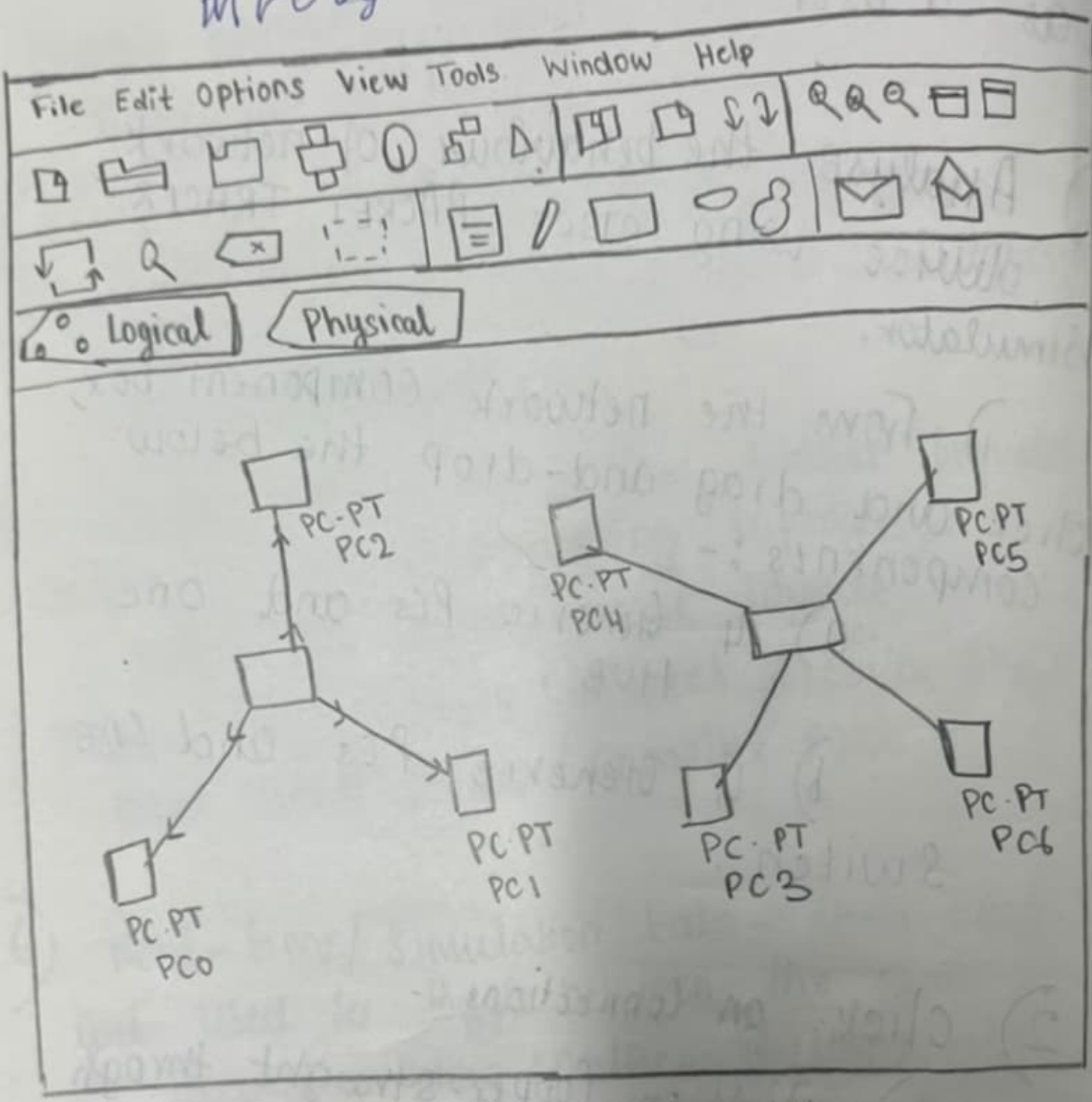
b) 4 Generic PCs and one Switch.

2) click on Connections:-

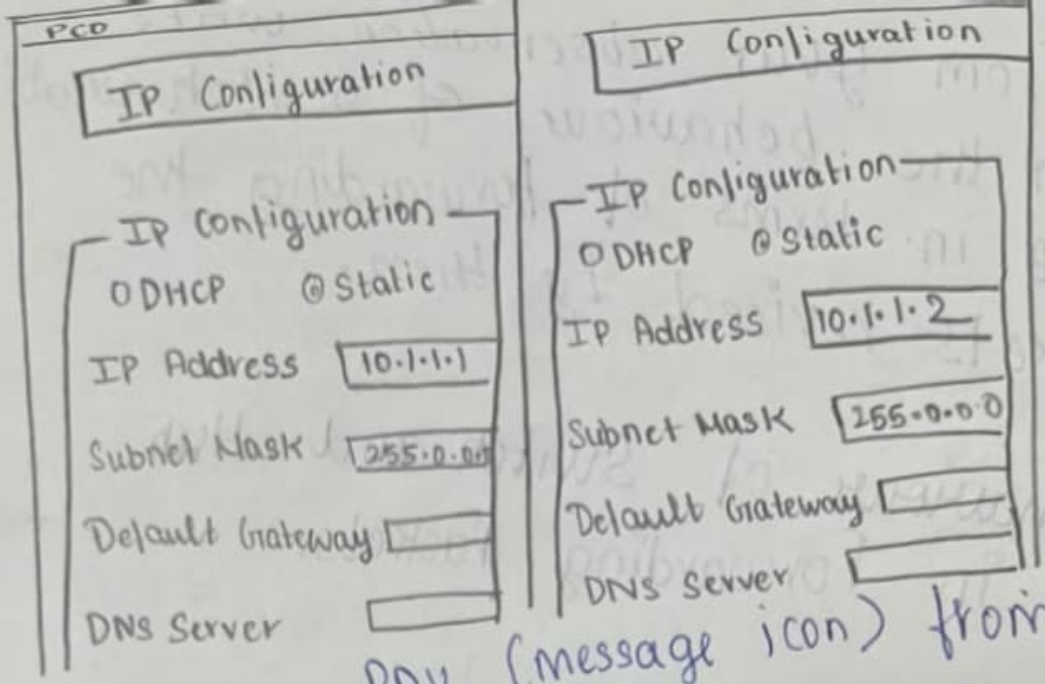
a) click on Copper-straight thrugh Cable,

b) Select on of the Pc and Connect it to HUB using the cable. the Link LED should glow in green, indicating that Link the is up. similarly, Connect remaining 3 PCs to the HUB.

c) Similarly connect 4 PCs to the switch using copper straight through cable.



3) click on the PC's connected to hub, go to Desktop tab, click on IP Configuration, and enter an IP address and subnet mask. Here, the default gateway and DNS server information is not needed as there are only two end devices in the network.



Click on the PDU (message icon) from the common tool bar,

a) Drag and Drop it one one of PC and then drop it on another PC connected to the HUB.

4) Observe the flow of PDU from source PC to destination PC by selecting the Realtime mode of simulation.

5) Repeat step #3 to step #5 for the PCs connected to the switch.

6) Observe how HUB and switch are forwarding the PDU and write your observation and conclusion about the behaviour of switch and HUB.

Student observation :-

a) From your observation write down the behaviour of switch and HUB in terms of forwarding the packets reviced by them.

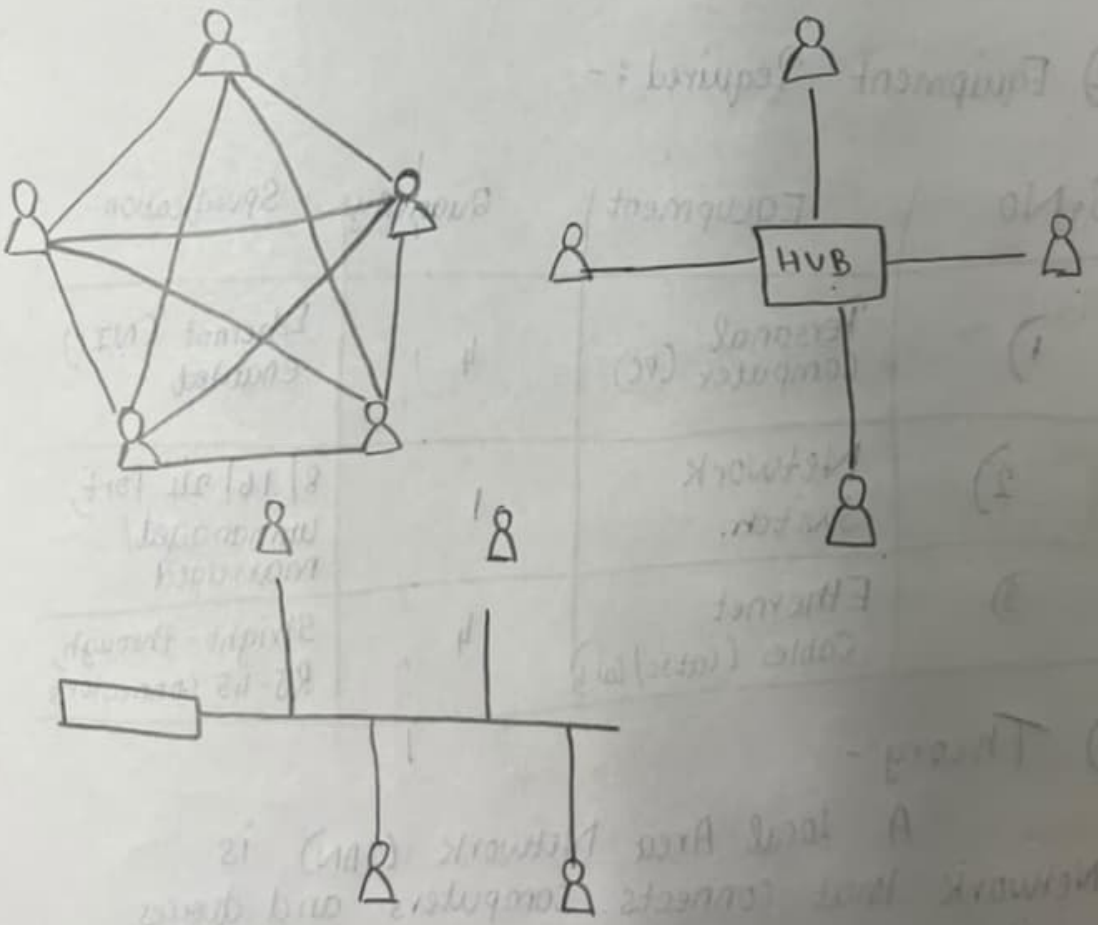
Behaviour of Switch and Hub in Forwarding Packets.

Hub	Switch
→ A Hub forwards the recieved data to all connected devices.	→ Inspects the destination address of each packet.
→ It copies incoming data to every single Port.	→ Forwards data only to the port of the target device.
→ Sends packet to all devices regardless of the target.	→ Reduces Network traffic and collisions.
→ Creates heavy Network traffic.	→ Keeps a table of connected device address.
→ Increase the chance of data collision.	

b) Find out the network Topology implemented in your college and label that topology in your observation book.

→ Colleges use Hiearchical Star Topology also known as Tree topology to connect various departments, computer labs, and main blocks to a single data center.

→ Network Topologies



Date:- 10/09/2026 PRACTICAL - 4

Setting up a local Area Network (LAN)

1) Aim :-

To design, set up, and configure a local Area Network (LAN) connecting four computers through a Network switch, assign static IP address to all devices, verify connectivity using the Ping command, and demonstrate resource/data sharing across the network.

2) Equipment Required :-

S.No	Equipment	Quantity	Specification
1)	Personal Computer (PC)	4	Ethernet (NIC) enabled
2)	Network Switch.	1	8/16/24 Port, unmanaged/ managed
3)	Ethernet Cables (Cat5e/Cat6)	4	Stright-through, RJ-45 connectors

3) Theory :-

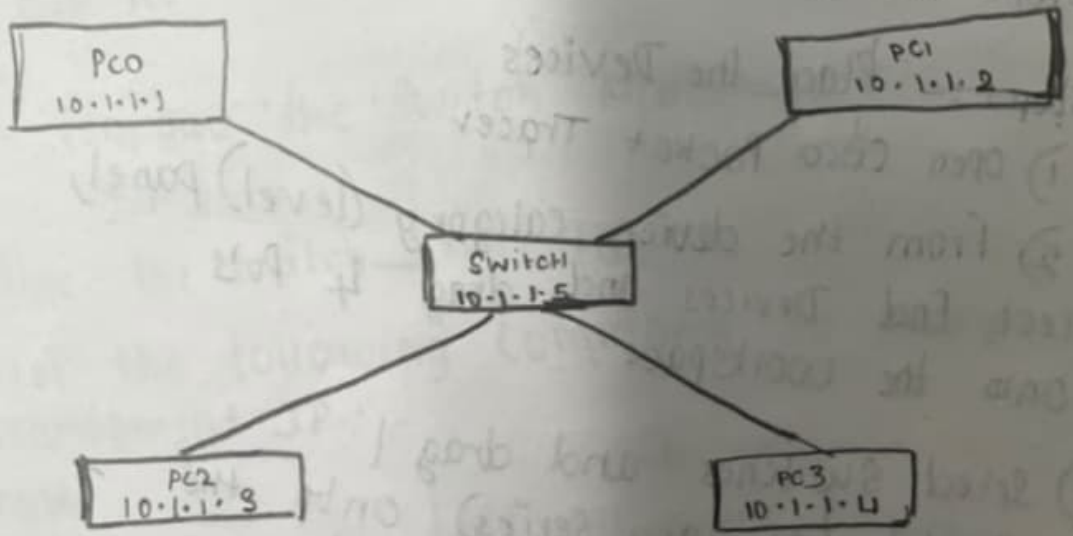
A local Area Network (LAN) is Network that connects computers and devices within a limited georgphical area such as a home, office / laboratory, allowing them to share resources such as files, Printers, and internet connections.

A star topology is used in this exp/ where All devices (PC's) are individually connected to a central network switch. the switch operates at layer 2 of the OSI model and forwards data frames between connected devices based on their MAC address. Every device on a LAN must be assigned an IP address within the same network range to be able to communicate with one another.

4) Network Diagram :-

the LAN was implemented using a star topology as shown below, with 4 PC's connected indivually to a central switch.

Star topology LAN (4PC's + Switch



5) Ip Address Configuration Table:-

Device	Ip Address	Subnet Mask	Connected To
PC0	10.1.1.1	255.0.0.0	Switch port Fa0/1
PC1	10.1.1.2	255.0.0.0	Switch port Fa0/2
PC2	10.1.1.3	255.0.0.0	Switch Port Fa0/3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa0/4
~~PC4~~ Switch (Management)	10.1.1.5	255.0.0.0	VLAN 1 Interface

6) Implementation in Cisco Packet Tracker

Since physical hardware was limited, the LAN was also implemented and verified using Cisco Packet Tracer simulation software, following the steps below.

Step 1:- Place the Devices

1) Open Cisco Packet Tracer

2) From the device category (level) panel, Select End Devices and drag 4 PC's onto the workspace.

3) Select Switches and drag 1 Switch (eg. 2960 series) onto the workspace.

Step 2:- Cable the Devices:-

4) Click Connections → Select Copper Straight-through cable.

5) Click each PC, then click the switch, and select an available Fastethrnet Port to complete each connection.

6) Repeat for all 4 PC's. Green link lights confirms a successful physical connections.

Step 3:- Assign Ip address to PC's

7) Click a PC → Desktop tab
↓
IP Configuration

8) Enter the static IP Address and Subnet mask as Per the IP table for each PC.

Step 4:- Configure the Switch Manangent IP.

9) Click the switch → CLI tab

10) Enter the following commands to assign a managment IP:-

```
enable
configure terminal
interface vlan 1
ip address 10.1.1.5 255.0.0.0
exit
```

Step 5 :- Verify Connectivity using Ping :-

11) click a PC ⟶ Desktop tab
↓
Command Prompt

12) Use the ping command to test connectivity to the other PC's, eg:-
Ping 10.1.1.2.

13) Successful "Reply from..." message confirm connectivity

Step 6 :- Demonstrate Data sharing (Server Simulation).

14) Add a server device
(End Devices ⟶ Server-PT) and connect it to the switch.

15) Assign it an IP address
(Eg :- 10.1.1.10, Subnet mask 255.0.0.0) via its Desktop ⟶ IP Configuration tab.

16) Open the server's service tab and enable the HTTP service.

17) From another PC, open the web browser and enter the server's IP

address to load the page, confirming successful data transfer accross the LAN.

Step 7: Observe Packet Flow in Simulation Mode.

19) Switch to simulation mode (bottom-right of the work space)

19) Send a ping or web request between two devices and use the Play/capture controls to step through as it travels hop-by-hop the packet as it travels hop-by-hop through the switch, observing MAC-address-based forwarding.

7. Observation

All four PCs were successfully connected to the switch in a star topology and assigned unique static IP addresses within the same subnet. (10.0.0.0/255.0.0.0). Ping tests between all devices returned succesful replies conforming end-to-end connectivity. the shared folder/server resource was successfully accessed from other computers on the network.

AIM- Experiments on Packet capture tool : Wireshark.

Packet sniffer

→ Sniffs messages being sent/recieved by your computer.

→ Store and display the contents of the various protocol fields in the messages.

→ Passive Program

- Never sends packets itself
- No packets addressed to it
- recieves a copy of all packets (sent / recieved)

Packet sniffer structure Diagnostic Tools

→ Tcpdump
- Eg :- tcpdump - enx host 10.129.41.2 -w exe3.out

→ Wireshark
- wireshark -r exe3.out

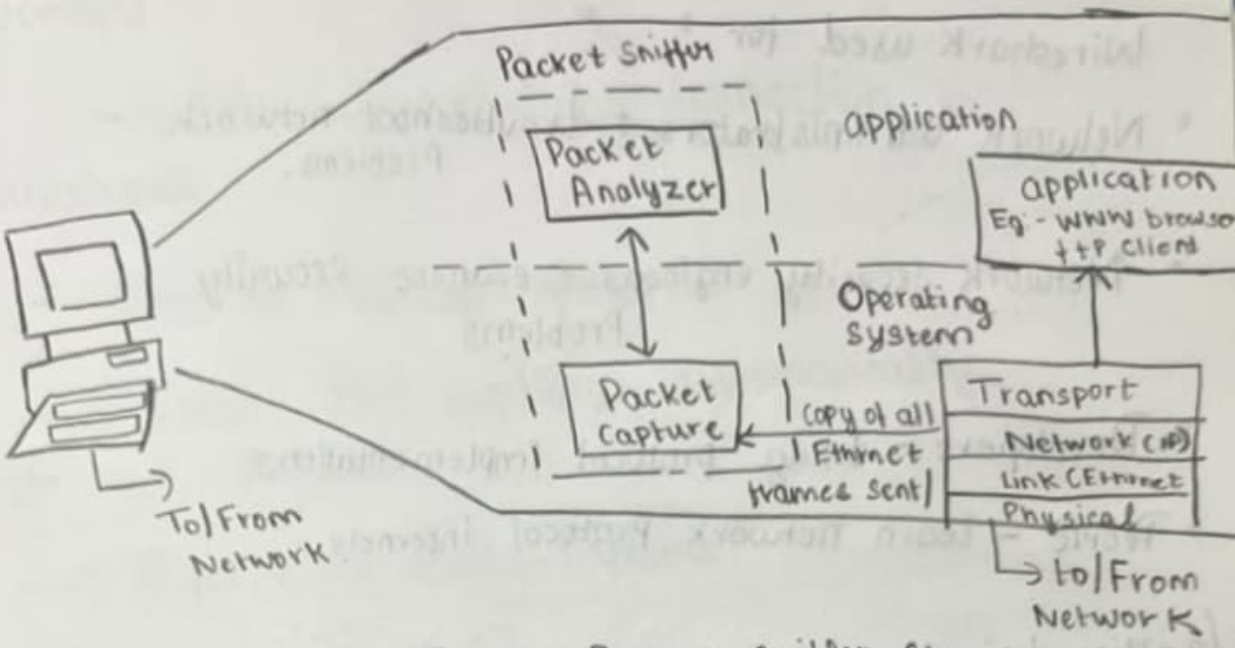


Fig 1:- Packet Sniffer Structure.

DESCRIPTION:-

WIRESHARK

Wireshark, a network analysis tool formerly known as Ethereal, captures packets in real time and display them in human-readable format. Wireshark includes filters, color coding, and other features that let you dig deep into network traffic and inspect indival packets. You can use Wireshark to inspect a suspecious program's network traffic, analyze the traffic flow on your network, or troubleshoot network problems.

What can we do with Wireshark:

- Capture network traffic
- Decode Packet Protocols using dissectors
- Define filters - capture and display
- Watch smart statistics
- Analyze Problems
- Interactively Browse that traffic

Wireshark used for :

- Network administrators : troubleshoot network Problems.
- Network security engineers : examine security Problems
- Developers : debug protocol implementations
- People - learn network protocol internels

Getting Wireshark

Wireshark can be downloaded for Windows (or) MacOs from it's official website. For Linux (or) another - UNIX like System, Wireshark will be found in its package repositeries. For Ubuntu, Wireshark will be found in Ubutun Software center.

Capturing Packets

After downloading and installing Wireshark launch it and double click the name of the network interface under Capture to start capturing packets on that interface.

Procedure :-

→ Select Local Area Connection in Wireshark.

→ Go to Capture option

→ Select Stop capture automatically After 100 packets.

→ Then click start capture

Output :-

1) Create a filter to display only TCP/UDP Packets, inspect the Packets and Provide the flow graph

Procedure :-

* Select Local Area Connection in wireshark
* Go to Capture option
* Select stop capture automatically after 100 packets.
* then click start capture
* Search TCP Packets in Search bar
* To see flow graph click statistics Flow graph.
* Save the packets.

2) Create a Filter to display only ARP Packets and inspect the Packets.

Procedure.

→ Go to Capture option

→ Select Stop capture automatically After 100 packets.

→ then click start capture

→ Search ARP Packets in search bar

→ Save the packets

3) Create a Filter to display only DNS Packets and provide the flow graphs Procedure.

→ Go to Capture option

→ Select stop capture Automatically After 100 Packets.

→ Then click start capture

→ Search DNS packets in search bar

→ To see flow graph click Statistic Flow graph

→ Save the Packet

4) Create a Filter to display only HTTP Packets and inspect the packets

Procedure.

→ Select Local Area Connection in Wireshark

→ Go to capture option

→ Select stop capture automatically after 100 packets

→ then click start capture

→ search HTTP packets in search bar.

→ save the Packets.

5) Create a Filter to display only IP/ICMP Packets and inspect the packets. Procedure

→ Select Local Area Connection in Wireshark

→ Go to Capture option

→ Select stop capture automatically after 100 packets.

→ then click start capture

→ Search ICMP | IP packets in search bar.

→ save the Packets.

6) Create a Filter to display only DHCP packets and inspect the packets.

Procedure:

→ Select Local Area Connection in Wireshark.

→ Go to Capture Option

→ then click start capture

→ search DHCP Packets in search bar.

→ Save the ~~tora~~ Packets.

Practical - 6

Aim:- Write a program to implement error detection and correction using HAMMING Code concept. Make a test run to input data stream and verify error correction feature.

Error Correction at Data Link Layer:

Hamming Code is a set of error-correction codes that can be used to detect and correct the errors that can occur when the data is transmitted from the sender to reciever.

Create Sender program with below features.

1) Input to sender file shold be a text of any length. Program should convert the text to binary.
2) Apply hamming Code concept on the binary data and add redundant bits to it.
3) Save this output in a file called channel.

Create a reciever program with below features.

1) Reciever program should read the input from channel file.
2) Apply hamming code on binary data to check for errors
3) If there is an error, display the position of the error.

4) Else remove the redundant bits and convert the binary data to ascii and display the output :

Student observation :-

Code :- sender Program :-

```
text = input ("Enter text:")
data = ""
for ch in text :
        data += format (ord (ch), "08b")
m = len (data)
r = 0
while 2 ** r < m + r + 1 :
        r += 1
n = m + r
code = [0] * (n+1)
j = 0
for i in range (1, n+1) :
        if i & (i-1) :
                code [i] = int (data [j])
                j += 1
for i in range (r) :
        P = 2 ** i
        Parity = 0
        for j in range (1, n+1) :
                if j & P :
                        Parity ^= code [j]
        code [P] = Parity
hamming = ""
```

```
for i in range (1, n+1):
    hamming += str (code [i])
With open ("channel.txt", "w") as F:
    f.write (hamming)
Print ("Binary Data :", data)
Print ("Hamming Code :", hamming)
Print ("Data saved to channel.txt")
```

Code :- Reciever Program :-

```
With open ("channel.txt", "r") as F:
    data = f.read().strip()
code = [0] + [int (x) for x in data]
n = len (data)
r = 0
While 2**r < n+1:
    r += 1

error = 0
for i in range (r):
    p = 2**i
    Parity = 0
    for j in range (1, n+1):
        if parity:
            error += p

if error != 0:
    Print ("Error detected at partion:",
        error)
    code [error] ^= 1
    Print ("Error Corrected")

result = ""
```

```
for i in range (1, n+1):
    if i & (i-1):
        result += str (code[i])
output = ""
for i in range (0, len(result), 8):
    binary = result [i : i+8]
    output += chr (int(binary, 2))
Print ("output:", output)
```

I/P:-

```
Enter text: Hellow
Binary data: 01001 00001 00010 10100 11...
Hamming code: 0000 1001 0010 0000 1101 001...
Data saved to channel.txt
```

O/P:-

```
No error detected
O/P:- Hello
```

Student observation:- Practical - 7.

Sender program Code :-

```
import time
msg = input ("Enter text:")
w = int(input ("Enter window size:"))
frames = []
for i in range (len(msg)):
    frames.append([i+1, msg[i]])

start = 0
while start < len(frames):
    end = min(start + w, len(frames))
    F = open ("Sender_Buffer.txt", "w")
    Print ("\n Frames sent:")
    for i in range (start, end):
        Print ("Frame no:", frames[i][0],
               "DATA:", frames[i][1])
        F.Write (str(frames[i][0] + " " +
        frames[i][i])
    F.close()
    Print ("Waiting for acknowledment...")
    time.sleep(3)
try:
    F = open ("Reciever_Buff.txt", "r")
    ack = int(f.Read().strip())
    F.close()
except:
    print ("No ACK recieved")
    continue
Print ("Recieved ACK:", ack)
```

```
if ack == end:
    Print ("Ack correct . Sending next window")
    start = end

else:
    Print("Nack recieved")
    Print("Resending Frames ...")
    time.sleep(2)
Print("\n All frames transimitted")
Print("Message :", msg)
```

: Reciever Program Code :-

```
import time
expected = 1
While True:
    try:
        F = open("sender_Buffer.txt", "r")
        lines = F.readlines()
        F.close()
    expect :
        print("Sender buffer not found")
        time.sleep(2)
        continue
    error = False
    Print("\n Forms recieved!")
```

```
for line in lines:
        Parts = line.split()
        data = parts[1]
        Print("Frame no:", frame, "DATA:", data)
        if frame != expected
            error = True
            break
        expected += 1
F = open("Reciever_Buffer.txt", "w")
if error:
        Print("wrong frame recieved")
        Print("Sending NACK:", expected)
        F.write(str(expected))
F.close()
time.sleep(1)
if expected > 1000:
        break
```

Expected Output :-

```
Enter file Text : HELLO
Enter window size : 3
Frames sent :
Frame no : 1 DATA : H
Frame no : 2 DATA : E
Frame no : 3 DATA : L
Waiting for acknowledgement
```

Recieved ACK : 4
ACK correct : Sendind next window
Frames sent :
Frame no : 4 DATA : L
Frame no : 5 DATA : O
Waiting for acknowledegment
Recieved ACK : 6
ACK Correct sending next window
All Frames Transimitted
Message : HELLO

Practical-8

Nmap Live Host Discovery ARP, ICMP and TCP/UDP Ping Scans - Student Lab Guide

TrackMe Room : Nmap Live Host Discovery | https://tryhackme.com/room/nmap01

Aim:-

To discover Live hosts on a network using Nmap host-discovery scans - ARP, ICMP and TCP/UDP Ping Scan and to understand which Protocol operates at which layer of TCP/IP model, and when each one Applies.

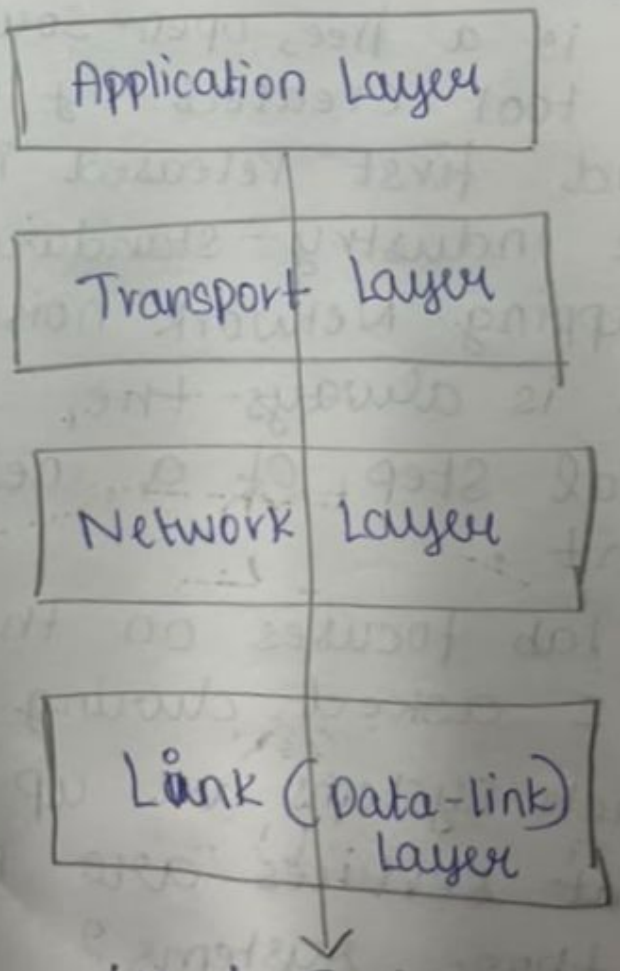
1. Introduction:-

Nmap is a free, open-source Network Scanning tool created by Gordon Lyon and first released in 1997. It is the industry-standard tool for mapping Network noise - so host discovery is always the first Practical step of a network assessment.

this lab focuses on the first two questions asked during reconnaissance

1. which systems are up?
2. What services are running on those systems?

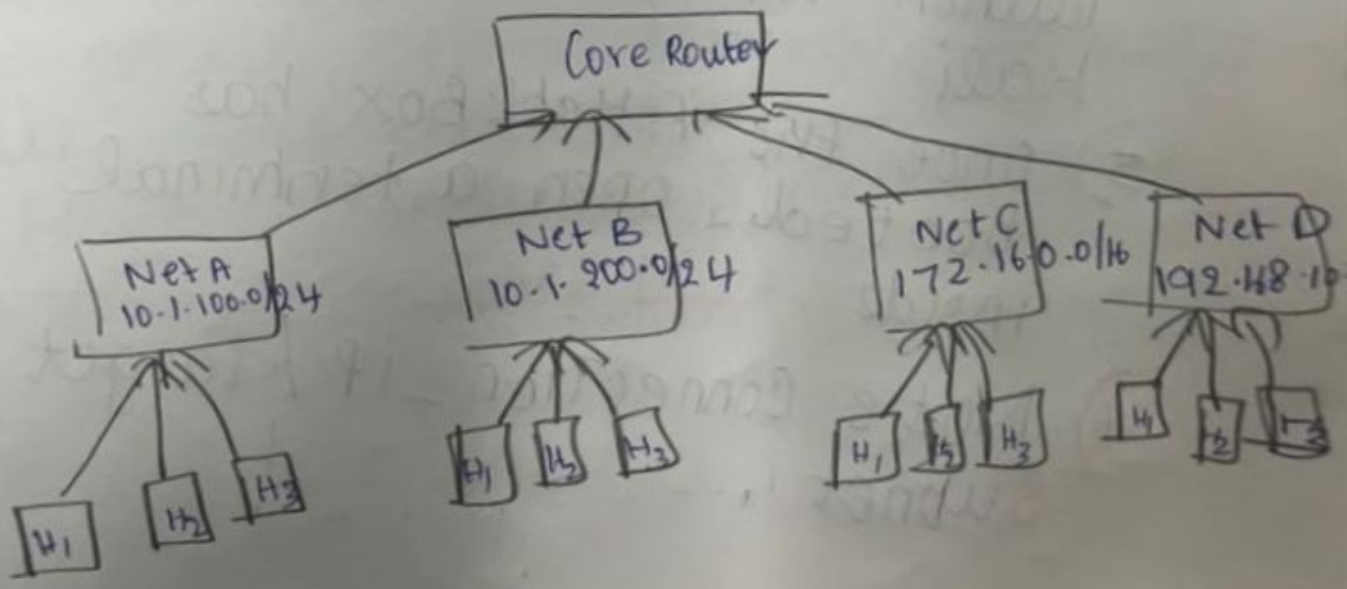
- ARP (Link Layer) - for hosts on the same local subnet
- ICMP (Network Layer) - echo, time stamp, and address-mask requests
- TCP - (Transport Layer) SYN and ACK ping probes
- UDP (Transport layer) relies on an ICMP "port Unreachable" reply.



Used First when the target is reachable at that layer.

Network segment and Sub Networks :-
Networ segment is a group of computers connected by a shared physical medium an Ethernet switch (or) a Wi-Fi access Point. A subnetwork is a logical grouping :- one or more segments that share the same IP address range and connect to wider network through the same router. a segment is a physical connection, subnet is a logical one.

A /24 Subnet holds roughly 250 usable hosts, a /16 Subnet holds roughly 65,000 hosts. firewall may sit btw segments and enforce policy on traffic that cross them



Target Specification in Nmap

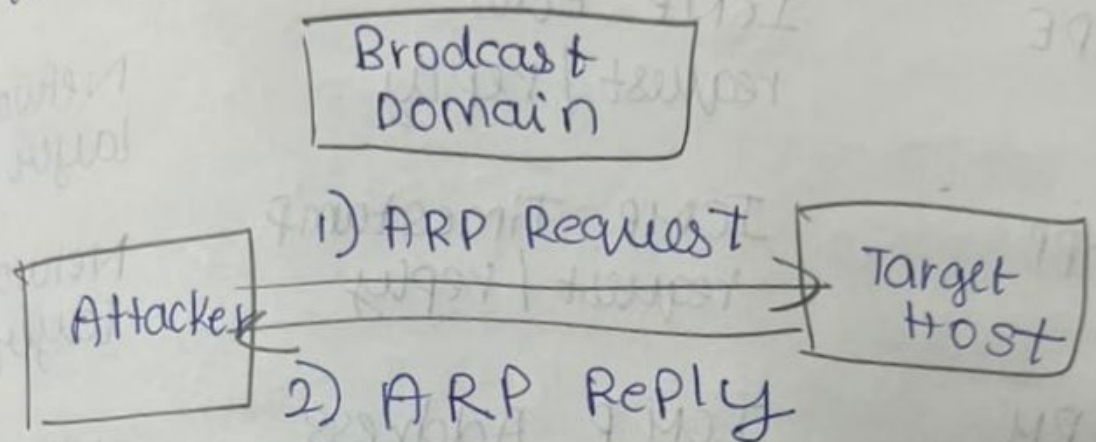
- A list : nmap 10.10.1.5 scanme.nmap.org example.com
- A range : nmap 10.11.12.15-20
- A Subnet / CIDR : 10.10.1.0/24
- File of targets : nmap -iL list_of_hosts.txt

Lab set up on TryHackMe

1) Open the room
2) Read Task 1 and Task 2
3) In Task 4 click Start to deploy the target network.
4) Click start AttackBox to launch the browser based Kali
5) Once the AttackBox has booted, open a terminal inside.
6) Note Connection_ip / target Subnet.

Host Discovery :-

1. ARP on a local Ethernet (or) wifi network, communication requires the destination's MAC address.



$ sudo nmap -PR -sn 10.200.6.0/24

O/P:-

Starting Nmap 7.80
Nmap scan report for 10.200.6.1
Host is up
Nmap done: 256 IP address
scannend in 10.51 sec.

2. ICMP Host Discovery :-

ICMP offers several requests Pair that can reveal a live Host. Windows system and Many firewall block.

-SN	Host-discovery only: skip port scanning	All
-PR	ARP scan	Link Layer
-PE	ICMP Echo request/reply	Network layer
-PP	ICMP Timestamp request/reply	Network layer
-PM	ICMP Address Mask req/reply	NL
-PS <port>	TCP SYN Ping	Transport layer
-PA <ports>	TCP ACK Ping	Transport layer
-PU <ports>	UDP Ping	Transport layer
-n	Disable reverse DNS resolution	—
-R	Force Reverse -DNS lookup	—
-iL <file>	Read target List from a file	

Step-by-step lab checklist:-

1) Log into TryHackMe and open the "Nmap Live Host Discovery"
2) Work through Tasks 1-3 and answer the simulator based questions.
3) Deploy the lab Network in Task 4 and start AttackBox
4) Run an ARP Scan: sudo nmap -PR -sn
5) Run an ICMP Timestamp and ICMP Address Mask (-PP) and (-PM) scans.
6) Run ICMP echo scan
7) Run a TCP SYN Ping (-PS) TCP ACK Ping
8) optionally capture traffic with tcpdump.
9) Try nmap -sL <subnet>
10) complete the room's knowledge-check questions in Task 5-9.

PRACTICAL

AIM: Implementation of SUBNETTING in CISCOPACKET TRACER Simulator.

Classless IP subnetting. Classless is a technique that allows for more effective use of IP addresses by allowing for submet masks that are not just the default marks for each IP class. This means we can divide our IP address space into smaller subnets, which can be useful when we have a limited number of IP addresses but need to create multiple networks

CREATING A NETWORK TOPOLOGY:

The I Step in implementing IP Subnetting is to create a network topology in Packet tracer. To create a network topology select "New" in the top left corner, then select "Network" and "Generic". This will create a blank network topology that we can use to add new

device

Adding the device:

Here we will be adding routers, switches, and PCs. To add a device select the device from the bottom left corner and drag it onto the network topology.

Subnetting:-

To subnet the network address of 192.168.1.0/24 to provide enough space for at least 5 addresses for each device, the device, and the router we can use a /27 subnet mask.

The IP addressing for the network shown in the topology can be as follows

- Router R1:
  - Gigabit Ethernet 0/0: 192.168.1.1
  - Gigabit Ethernet 0/1: 192.168.2.1
- Switch S1:
  - FastEthernet 0/1: 192.168.1.0/27

device

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- Router R1 :
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  - Gigabit Ethernet 0/1 : 192.168.2.1
- Switch S1 :
  - FastEthernet 0/1 : 192.168.1.0/27

- PC1 : 192.168.1.11
- PC2 : 192.168.1.12
- PC3 : 192.168.1.13
- PC4 : 192.168.1.14
- PC5 : 192.168.1.15
- FastEthernet 0/2 : 192.168.2.0/29
- PC1 : 192.168.2.11
- PC2 : 192.168.2.12
- PC3 : 192.168.2.13
- PC4 : 192.168.2.14
- PC5 : 192.168.2.15
- Router R2 :
  - FastEthernet 0/0 : 192.168.3.1
  - fastEthernet 0/1 : 192.168.4.1
  - Switch S2 :
    - FastEthernet 0/1 : 192.168.3.0/29
    - PC1 : 192.168.3.11
    - PC2 : 192.168.3.12
    - PC3 : 192.168.3.13
    - PC4 : 192.168.3.14
    - fastEthernet 0/2 : 192.168.4.0

PC1: 192.168.4.11
PC2: 192.168.4.12
PC3: 192.168.4.13
PC4: 192.168.4.14
PC5: 192.168.4.15

CONFIGURING A FITE DEVICE:

We can start configuring them. We have to Right click on the router and select "CLI". In that enter

enable
configure terminal
interface fastEthernet 0/0
ip address [IP address] [subnet mask]
no shutdown
exit
interface FastEthernet 0/1
ip address [IP address] [subnet mask]
no shutdown
exit

Testing the Network: Open a command prompt on each PC and try to ping the other PC. If the ping is successful then the network is functioning properly. We can also use the "ping" command to test connectivity between the Router and PC's

PRACTICAL ..

Aim a) Interneworking with routers in CISCO PACKET TRACER Simulator

b) Design and configure a simple internetwork using a router.

In this network a router and 2 PCs are used. Computers are connected with routers using a copper straight-through cable. After forming the network to check network connectivity a simple PDU is transferred from PC0 to PC1.

Procedure:

Step 1 : [Configuring Router 1]:

1) Select the router to select open CLI
2) Press Enter to start Configuring.
3) Type enable to activate the privileged mode.

Router 1 Command Line Interface.

Router > enable

Router # config t

Enter configuration commands, one per line. End with CNTL/Z.

Router (config) # interface FastEthernet 0/0

Router (config-if)# ip address 198.168.10.1
255.255.255.0 Router (config-if)# no
shutdown

Router (config-if)# Interface Gigabit Ethernet 0/0
%LINK-5-CHANGED: • Changed
State to up Router(config-if)# Interface
fastEthernet 0/1
Router(config-if)# ip address 192.168.20.1
255.255.255.0 Router (config-if)# noshutdown

Step 2 (Configuring PC's)

1) Assign IP address to every PC in the network.
2) select the PC, go to the desktop and select IP configuration and assign an IP address, Default gateway, subnet Mask
3) Assign the default gateway
PC0 as 192.168.10.1
4) PC1 as 192.168.20.1

Step 3:

1) connect fastEthernet 0 port of PC0 with Ethernet 0/0 port of Router using a copper straight-through cable

a) Connect FastEthernet0 port of PC1 with FastEthernet0/1 port of Router1 using a copper straight through Cable.

PRACTICAL 03

Aim: a) Design and configure an internetwork using wireless router, DHCP server and Internet cloud

Objectives:

Part 1: Build a simple Network as per Logical Topology workspace.

Part 2: Configure the Network Devices

Part 3: Test connectivity between Network Devices

Part 4: Save the file and close Packet Tracker

Part 1: Build a Simple Network as per Logical Topology workspace.

Part 1: Launch Packet Tracer.

Step 2: Launch Build the Topology. Add network to the device to the workspace

first choose a device type from the Device-Type Selection box. Then click on the desired device model from the device-specific selection box. Click a location in the workspace. If you want to cancel click cancel icon. You can click and drag a device from the device-specific selection box

b) Change display names of the network devices.

Click on device icon, then click on the config tab in the device configuration window. Type the name of the device

c) Add the physical cabling between devices on the work space

Using the device selection box, add the physical cabling between devices.

The PC will need a copper straight through cable to connect to the wireless router.

Part 2: Configure the Network devices

Step 1: Configure the wireless router

a) Click on the wireless Router icon to open the device configuration window.

b) Click on the GUI tab to view configuration options.

Next, Click on the wireless tab in the GUI, to view the wireless setting. Network Name (SSID). Here type the name "HomeNetwork"

b) Click on the save settings tab

Step 2: Configure the laptop.

Click on the Laptop icon. Select Physical tab.

Then remove the currently installed Ethernet copper module on the side of the Laptop and dragging it into MODULES

Once the wireless-N Notebook Adapter setting are done select the connect tab.

Step 3: Configure the PC.

Click the PC icon on the workspace and select Desktop tab. Then the IP

configuration tab. Select DHCP radio buttos. to receive an IPv4 from the wireless router. Close the IP configuration window

Step 4:

Configure the Internet cloud.

a) Install network modules if necessary.

Click on the internet cloud.

b) Identify the from and TO ports:

Click on the config tab. Click on cable and connections. Ethernet then click on the Add button. to add them

Step 5: Configure the Cisco.com Server

a) Click on the cisco.com select services tab. select DHCP and confirm a DHCP

- Click on to the DCHP service.
- On
- pool name: DHCP Pool
- Default gateway: 208.67.220.220
- DNS server. 208.67.220.220
- Starting IP address: 208.67.220.1
- Subnet Mask. 255.255.255.0
- Maximum number of users: 50

b). click on to run
- Name: Cisco.com
- Type: A Record
- Address: 208.67.220.220

c. Select the config tab
click on the settings in left pane.
- Select Static
- Gateway: 208.67.220.1
- DNS server: 208.67.220.220

d. Click on Fast Ethernet
- select static under IP configuration
- IP Address: 208:67.220.220
- Subnet mask: 255.255.255.0

Part 3: Verify connectivity
click on the command prompt icon
Refresh IP settings by issuing the command ipconfig/release and then ipconfig/renew the output should show the PC has an IP address in the 192.168.0.x range a subnet mask, a default gateway and DNS server address
b) Test Connectivity Cisco.com Server
From the command prompt, issue the command ping cisco.com, It may take a few seconds to return. 4 replies will be received

PRACTICAL 11

Aim: a) Simulate Static Routing Configuration. using CISCO PACKET TRACER

Static routes are the routes you manually add to the routers routing table. The process of adding is known as static routing.

setting up a practical lab

Create a packet tracer Lab.

Each network has two routes to reach. We will configure one route as the main router and another one as the backup route. If the link bandwidth of all routes is the same, we use the route that has the least number of routers as the main route.

If you manually want to select a route then, the router should add to the routing table you have to set AD Value.

ip route 30.0.0.0 255.0.0.0 20.0.0.2 10

ip route 30.0.0.0 255.0.0.0 40.0.0.2 20

creating, adding, verifying static routes
Routers automatically learn their connected networks.

Router 0 requirements

- create two routes for network 30.0.0.0/8 and configure the first route (via-Router 1) as main route and the second route (via-Router2) as backup route.
- create two routes for the host 30.0.0.100/8 and configure first route (via router1) as the main route and the second route (via-Router 1) as a backup route
- create two routes for network 50.0.0.0/8 and configure the first route (via-Router 2) and the second route (via Router 1) as backup
- Verify the router adds only main routes to the routing

Route 1 requirements

1. Create two route for network 10.0.0.0/8 and configure the first route (via Router 0) as the main route and the second route (via Router 1) as a backup
2. Create two routes for network 40.0.0.0/8 and configure the first route (via Router 0) as the main Route and the second Route (via Router 2) as backup
3. Verify the route adds only main routes to the routing table.

Router 2 configuration

Router > enable
Router # configure
terminal
Enter Configuration commands, one per li
End with CNTL/Z
Router(config)# ip route 10.0.0.0 255.0.0.0 4000
Router(config)# ip route 30.0.0.0 255.0.0.0
30.0.0.2 Router (config)# exit
Router # show ip route static
S 10.0.0.0/8 [1/0] via
40.0.0.1 S 30.0.0.0/8 [1/0]

Verifying Static routing

By sending ping request to a PC of network 30.00.0/8 and tracing the path they take to reach the network 30.0.0.0/8. For this you can use 'tracert' command. on 10.0.0.0/8. This commands send ping request to the destination host and track the path they take to reach the destination.

PROGRAM 11

Aim : b) Simulate RIP using CISCO Packet tracer

Assign IP address to PCs

Double click PC's and click Desktop menu and click IP configuration

Assign IP address.

Router> enable.

Router# configure

Terminal

Enter configuration commands, one per line End with CNTR/Z

Configure RIP routing Protocol

Configuration of RIP Protocol is much easier than you think. It requires only two steps to configure RIP routing

. Enable RIP routing protocol from the global configuration mode.

. Tell RIP routing protocol which networks you want to advertise.

Router 0

router rip command tell router to enable the RIP routing protocol.

network command allows us to specify the network which we want to advertise.

Router 1

Router 1 [config] # router rip

Router 1 [config-router) # network 192.168.1.248

Router 1 [config-router] # network 192.168.1.248

Router 2

Router 2 (config) # router rip

Router 2 (config-router) # network 20.0.0.0

Router 2 (config-router) # network 192.168.1.252

Router 2 (config-router) # network 192.168.1.248

RIP Protocol automatically manage all routes for us.

PRACTICAL 12

Aim: a) Implement echo client server using TCP/UDP sockets

Algorithm:

Input

output

Algorithm:-

A) TCP Echo Server

1) Start the Program.

2) Create a TCP socket using socket()

3) Bind the socket to an IP address and port using bind()

4) Listen for client connections using Listen()

5) Accept the client connection using accept()

6) Recieve a message from the client using rev()

7) Send the same message back to the client using send()

8) Close the connection.

9) Stop the Program

B) Tcp Echo Client:-

1) Start the program
2) Create a TCP Socket using socket()
3) connect the server using connect()
4) Enter a msg and send it to the server.
5) Recieve the echoed message from the server.
6) Display the recieved message.
7) close the socket
8) stop the program

C) UDP Echo Server:-

1) start the program
2) create a UDP socket using socket()
3) Bind the socket to an IP address and port.
4) Recieve message using recvfrom()
5) Send same message back using sendto()
6) Close the socket
7) Stop the program

Input:-

Enter message :- Hello Server

O/P :-

Server Side:-

Server started ...
Waiting for client ...
Recieved message : Hello Server
Echo message sent successfully.

Client Side:-

Connected to Server
Message Sent: Hello Server
Message Recieved : Hello Server.

Result:-

Thus, echo client Server application using TCP and UDP sockets was implemented Successfully, and msg sent by the client were echoed back by the server.

Aim :-

b) Implement chat client server using TCP/UDP sockets.

Algorithm :-

1) Start the program
2) import the required socket library
3) Create a server socket and a client socket.
4) Assign an IP address and Port no to the server.
5) TCP, establish connection between the client and server using connect() and accept().
6) In UDP, exchange message using sendto() and recfrom() without establishing a connection.
7) client sends message to server.
8) server recieves the message and displays it.
9) Server sends a reply to client.
10) Repeat the process until either side Sends "bye"
11) Close the socket and stop the program.

I/P:-
Message entered by client and server through keyboard.

o/p:-

Server started...
Waiting for client...
client connected.
client: Hello Server
server: Hello client
client: How are you?
server: I am fine
client: bye
connection closed

client Side:-

Connected to server
Client: Hello server
Server: Hello client
Client: How are you?
Server: I am fine.
client: bye
Connection closed.

Result:-
Thus, Chat application using TCP/UDP sockets was implemented successfully, and message were exchanged between clien and server.

Practical - 13

Aim:-
Implement your own Ping Program.

Algorithm:-

1) Start the program
2) Enter the destination IPA
3) Create a raw socket using ICMP Protocol.
4) Construct and send an ICMP Echo Request Packet.
5) Wait for an ICMP Echo reply from the destination
6) Calculate the Round trip time (RTT)
7) Display the reply and response time.
8) Repeat for the required number of Packets.
9) Close the socket and stop

Program:-

```
import Socket
import struct
import time
import os
```

```
def checksum(data):
    if len(data) % 2:
        data += b'\x00'
    total = sum((data[i] << 8)
for i in range(0, len(data), 2))
    while total >> 16:
        total = (total & 0xffff) +
(total >> 16)
    return ~total & 0xffff
def ping(host):
    ip = socket.gethostbyname(host)
    print("pinging", ip)
sock = socket.socket(socket.AF_INET,
                     socket.SOCK_RAW,
                     socket.IPPROTO)
    except socket.timeout:
        print("Requested timed
              out")
    time.sleep(1)
    sock.close()
host = input("Enter IP address:")
ping(host)
```

I/P :-

Enter IP address : 8.8.8.8

O/P :-

Pinging 8.8.8.8

Reply from 8.8.8.8 : time = 15.24 ms

Reply from 8.8.8.8 : time = 16.12 ms

Reply from 8.8.8.8 : time = 15.67 ms

Result :-

Thus, the ping program was implemented to test network connectivity using ICMP packets.

Practical

Packet sniffing using Raw sockets.

Aim:-

To write a program using raw sockets to capture and analyze network packets.

Algorithm:-

1) Start the program
2) Create a new socket to capture network packets.
3) Bind the socket to the network protocol
4) Recieve incoming Packets using the socket
5) Extract the Ethernet and IP address information
6) identify the source IP, destination IP, and protocol
7) Display the captured packet details
8) Repeat unit the user terminates the program

9) close the socket and stop

Program :-

```
import socket
s = socket . socket ( socket.
    AF_ PACKET, socket . SOCK_RAW
socket ntohs (3))
    while True:
        a = s.recfrom (65565)
        Print (a[0] .next())
```

Input :-

Network packets recieved by the system.

O/P :-

Packet sniffer started...
Displays captured packets in hexadecimal format.

Result :-

thus packet sniffing was implemented successfully using raw packets.